

COMBINATORIAL ALGEBRAIC TOPOLOGY
AND
APPLICATIONS TO DISTRIBUTED COMPUTING

Dmitry Feichtner-Kozlov

University of Bremen

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Lecture I: Simplicial methods in theoretical
distributed computing

Lecture II: Topology of protocol complexes

Lecture III: Mathematics of the Weak
Symmetry Breaking task

Lecture I: Simplicial methods in theoretical distributed computing

Plan of Lecture I:

1. Combinatorial Algebraic Topology
2. A primer in theoretical distributed computing
3. Simplicial models for tasks and executions

1. Combinatorial Algebraic Topology

What is ...

Combinatorial Algebraic Topology ?

Combinatorial Algebraic Topology is about computing invariants in Algebraic Topology

- for **combinatorial cell complexes**
- by **combinatorial means**.

Combinatorial cell complexes

locally = simple cell attachments

for example - simplicial or prodsimplicial complexes;

globally = cells are indexed by combinatorial objects

such as graphs, partitions, permutations, various combinations and enrichments of these;

taking boundary = combinatorial rule for the indexing objects

for example - removing vertices in graphs, merging blocks in partitions, relabeling, etc.

Combinatorial means

using matchings, orderings, labelings, et cetera, to simplify or to completely eliminate algebraic computations and topological deformations.

Theory

- Discrete structures derived as models for topological spaces.
- Many constructions of cell complexes with discrete structures as input.
- Using standard Algebraic Topology tools uncovers ties between discrete structures.
- Properties of discrete structures are distinguished by how well they behave with respect to Algebraic Topology.

Applications

- The study of discrete structures yields information about the topology of the original spaces and vice versa.

Hom Complexes

Definition.

Given two graphs G and H , the complex $\text{Hom}(G, H)$ is the complex of all multihomomorphisms from G to H .

In particular, the vertices of $\text{Hom}(G, H)$ are graph homomorphisms from G to H .
The cells are direct products of simplices: *prodsimplicial complexes*.

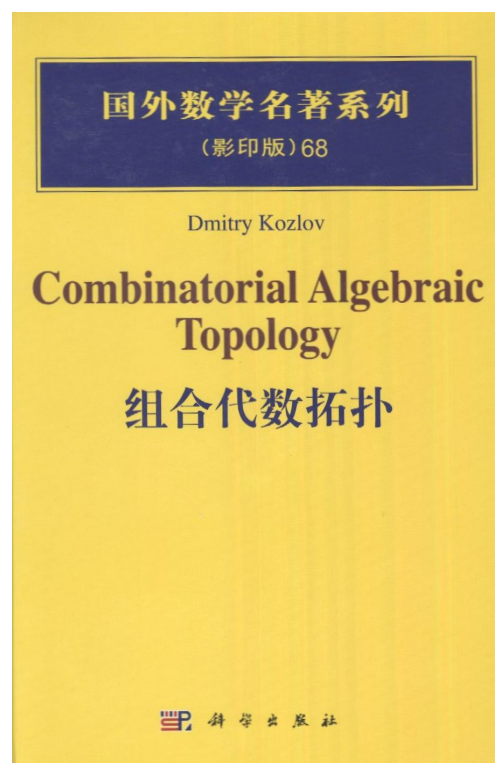
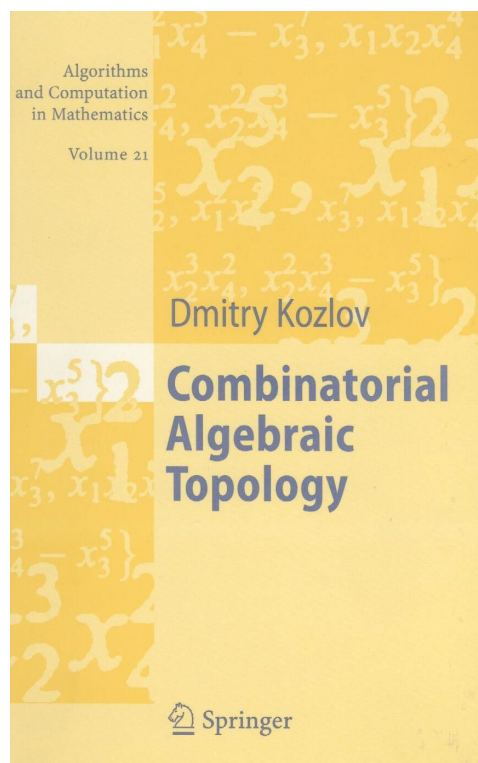
Theorem.

Babson-Kozlov 2007, Annals of Mathematics.

Assume $\text{Hom}(C_{2r+1}, G)$ is k -connected, then $\chi(G) \geq k + 4$.

Reference

D. Kozlov: *Combinatorial Algebraic Topology*,
405 pp., Springer-Verlag, 2008.



2: A primer in theoretical distributed computing

- **n processes performing a certain task**
- **choice of the model of communication**
 - message passing
 - read/write
 - immediate snapshot
 - read/write with other primitives
- **choice of possible failures**
 - crash failures
 - byzantine failures
- **choice of timing model**
 - asynchronous
 - synchronous
- **impossibility results**

One standard model of communication

- processes communicate via shared memory using two operations:
 - (1) **write** - process writes its state to the devoted register;
 - (2) **snapshot read** - process reads the entire shared memory in an atomic step;
- allow crash failures;
- all protocols are asynchronous and **wait-free**, meaning there is no upper bound on how long it takes to execute a step;
- all executions are **immediate snapshot**, meaning that at each step a group of processes becomes active, all processes in that group first write and then read together;
- all executions are **layered**, meaning that the execution is divided into rounds; in each round each non-faulty process reads and writes exactly once.

An example: assigning communication channels

- The processes use shared memory to coordinate among themselves the assignment of unique communication channels.
- One can show that two available communication channels are not enough for two processes.
- Two processes can pick unique communication channels among three available channels c_1 , c_2 , and c_3 using the following protocol:
 - (1) the process writes its id into shared memory, then it reads the entire memory;
 - (2) if the other process did not write its id yet - pick c_1 ;
 - (3) if the other process wrote its id - compare the ids and pick c_2 or c_3 .

Examples of tasks

- **Binary consensus:** Each process starts with a value 0 or 1 and eventually decides on 0 or 1 as its output value, subject to the following conditions:
 - the output value of all the processes should be the same;
 - if all the processes have the same initial value then they all must decide on that value.
- **k -set agreement:** Each process starts with an input value and eventually decides on an output value, subject to the following conditions:
 - at most k different values appear as output values of the processes;
 - only values which appear as input values are allowed to be picked as an output value.
- **Weak Symmetry Breaking:** Processes have no inputs (*an inputless task*) and need to pick outputs from $\{0, 1\}$, such that not all processes pick the same output; the processes are only allowed to compare their id's with each other.

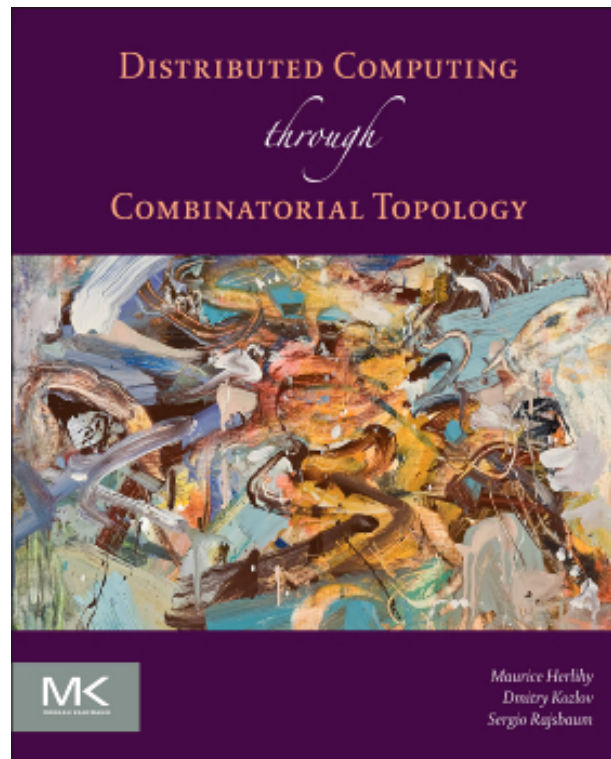
3: Simplicial models for tasks and executions

Reference

M. Herlihy, D. Kozlov, S. Rajsbaum:

Distributed Computing Through Combinatorial Topology,

319 pp., Elsevier / Morgan Kaufmann, 2014.



Introducing simplicial structure

- A simplicial complex of initial configurations: *input complex*
- A simplicial complex of final configurations: *output complex*
- A simplicial complex of all executions: *protocol complex*
- A carrier map from the input complex to the output complex: *task specification map*
- A carrier map from the input complex to the protocol complex: *execution map*
- A simplicial map from the protocol complex to the output complex: *decision map*
- A condition connecting task specification, decision, and execution maps

Binary consensus

Simplicial formulation for the binary consensus:

- The input complex $\mathcal{I}$ has vertices labeled (P, v) , where $P \in [n]$, $v \in \{0, 1\}$. The vertices $(P_0, v_0), \dots, (P_l, v_l)$ form a simplex iff $P_i \neq P_j$.
- The output complex $\mathcal{O}$ consists of two disjoint n -simplices; the first simplex has vertices labeled $(P, 0)$, and the second one has vertices labeled $(P, 1)$, for $P \in [n]$.
- The carrier map Δ is defined as follows. Take $\sigma = \{(P_0, v_0), \dots, (P_l, v_l)\} \in \mathcal{I}$.
 - If $v_0 = \dots = v_l = 0$, then $\Delta(\sigma)$ is the simplex $\{(P_0, 0), \dots, (P_l, 0)\}$;
 - If $v_0 = \dots = v_l = 1$, then $\Delta(\sigma)$ is the simplex $\{(P_0, 1), \dots, (P_l, 1)\}$;
 - else $\Delta(\sigma)$ is the union of the simplices $\{(P_0, 0), \dots, (P_l, 0)\}$ and $\{(P_0, 1), \dots, (P_l, 1)\}$.

Simplicial formalization of other standard tasks

- Consensus.
- k -set agreement.
- Weak Symmetry Breaking.

Simplicial formalization: notion of a task

Definition.

A **task** is a triple $(\mathcal{I}, \mathcal{O}, \Delta)$, where

- $\mathcal{I}$ is a rigid chromatic *input complex* colored by $[n]$ and labeled by V^{in} , such that each vertex is uniquely defined by its color together with its label;
- $\mathcal{O}$ is a rigid chromatic *output complex* colored by $[n]$ and labeled by V^{out} , such that each vertex is uniquely defined by its color together with its label;
- Δ is a chromatic carrier map from $\mathcal{I}$ to $\mathcal{O}$.

A chromatic abstract simplicial complex (K, χ) is called **rigid chromatic** if K is pure of dimension n , and χ is an $(n + 1)$ -coloring.

Assume K and L are chromatic simplicial complexes, and $\mathcal{M} : K \rightarrow 2^L$ is a carrier map. We call $\mathcal{M}$ **chromatic** if for all $\sigma \in K$ the simplicial complex $\mathcal{M}(\sigma)$ is pure of dimension $\dim \sigma$, and we have $\chi_K(\sigma) = \chi_L(\mathcal{M}(\sigma))$; where $\chi_L(\mathcal{M}(\sigma)) := \{\chi_L(v) \mid v \in V(\mathcal{M}(\sigma))\}$.

Simplicial formalization: protocol complexes

Definition.

A **protocol** for $n + 1$ processes is a triple $(\mathcal{I}, \mathcal{P}, \mathcal{E})$ where

- $\mathcal{I}$ is a rigid chromatic simplicial complex colored with elements of $[n]$ and labeled with V^{in} , such that each vertex is uniquely defined by (color,label);
- $\mathcal{P}$ is a rigid chromatic simplicial complex colored with elements of $[n]$ and labeled with Views_{n+1} , such that each vertex is uniquely defined by (color,label);
- $\mathcal{E} : \mathcal{I} \rightarrow 2^{\mathcal{P}}$ is a chromatic intersection-preserving carrier map, such that $\mathcal{P} = \cup_{\sigma \in \mathcal{I}} \mathcal{E}(\sigma)$, where intersection-preserving means that for all $\sigma, \tau \in \mathcal{I}$ we have

$$\mathcal{E}(\sigma \cap \tau) = \mathcal{E}(\sigma) \cap \mathcal{E}(\tau).$$

Definition.

Assume we are given a task $(\mathcal{I}, \mathcal{O}, \Delta)$ for $n + 1$ processes, and a protocol $(\mathcal{I}, \mathcal{P}, \mathcal{E})$. We say that this protocol **solves** this task if there exists a chromatic simplicial map $\delta : \mathcal{P} \rightarrow \mathcal{O}$, called **decision map**, satisfying

$$\delta(\mathcal{E}(\sigma)) \subseteq \Delta(\sigma),$$

for all $\sigma \in \mathcal{I}$.

Topological characterization of solvability of colored tasks

Theorem. Herlihy et al.

A task $(\mathcal{I}, \mathcal{O}, \Delta)$ has a wait-free layered immediate snapshot protocol if and only if there exists t and a (color-preserving) simplicial map

$$\mu : \text{Ch}^t(\mathcal{I}) \rightarrow \mathcal{O}$$

carried by Δ .

Sperner's Lemma.

Assume we are given

- a simplicial subdivision of Δ^n , called $\text{Div } \Delta^n$;
- an $[n]$ -coloring of the vertices of $\text{Div } \Delta^n$, such that for every vertex v only the colors of the carrier of v may be used.

Then $\text{Div } \Delta^n$ has a properly colored n -simplex.

Sperner's Lemma $\Rightarrow$ k -set agreement is not solvable.

SUMMARY

- Simplicial methods are useful in theoretical distributed computing.
- Recent work provides rigorous foundation for this interdisciplinary area.
- Protocol complexes are configuration spaces encoding all possible executions of protocols.
- Combinatorial topology of protocol complexes is sophisticated, contains answers to distributed computing questions.

References:

Attiya, Welch, *Distributed Computing*

M. Herlihy, D.N. Kozlov, S. Rajsbaum, *Distributed Computing through Combinatorial Topology*, 1st Edition, Morgan Kaufmann / Elsevier, 2014, 336 pp.

M. Herlihy, Shavit *Asynchronous*

D.N. Kozlov, *Combinatorial Algebraic Topology*

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Lecture II: Topology of protocol complexes

Plan of Lecture II:

1. Layered immediate snapshot protocol complexes
2. Immediate snapshot protocol complexes
3. Snapshot protocol complexes

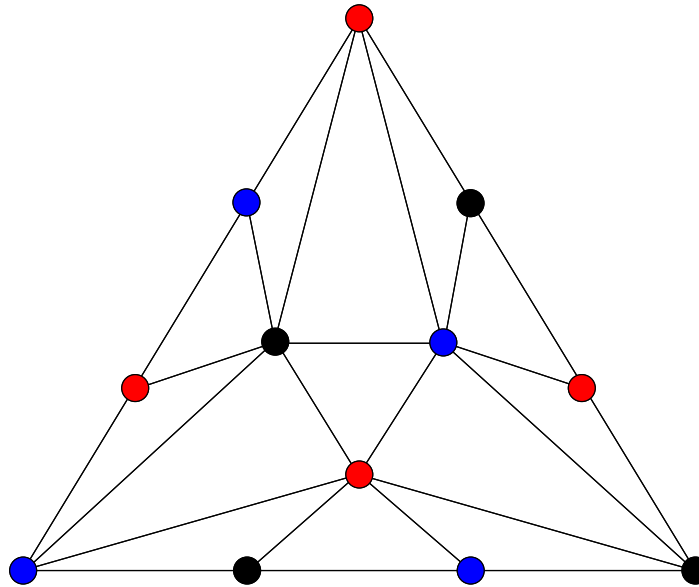
1. Layered immediate snapshot protocol complexes

Chromatic subdivision of a simplex

Example.

The protocol complex in the layered immediate snapshot model, where each of the $n + 1$ processes acts exactly once, is a chromatic subdivision of an n -simplex, which we call the **standard chromatic subdivision**.

Notation: $\text{Ch } \Delta^n$.



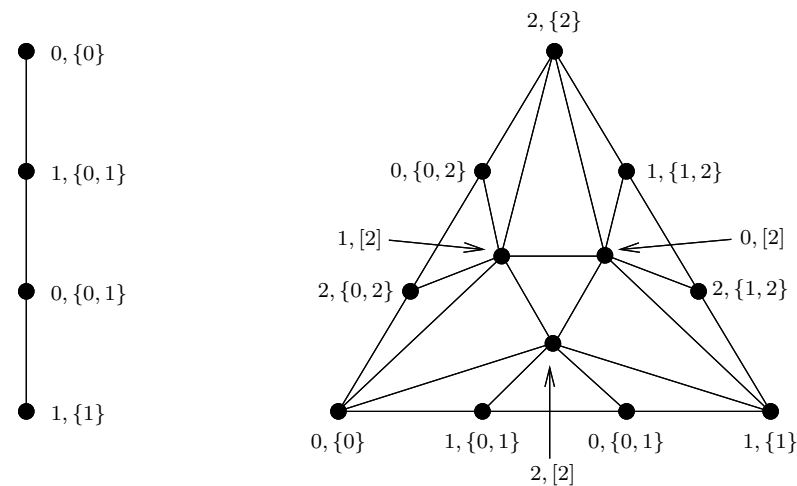
The protocol complex of the standard 1-round wait-free layered immediate snapshot protocol for 3 processes.

Topology of the layered immediate snapshot protocol complexes I

Theorem.

Protocol complexes for the standard 1-round wait-free layered immediate snapshot protocols for $n + 1$ processes are equivariantly collapsible chromatic subdivisions of an n -simplex.

Notation: $\text{Ch } \Delta^n$.

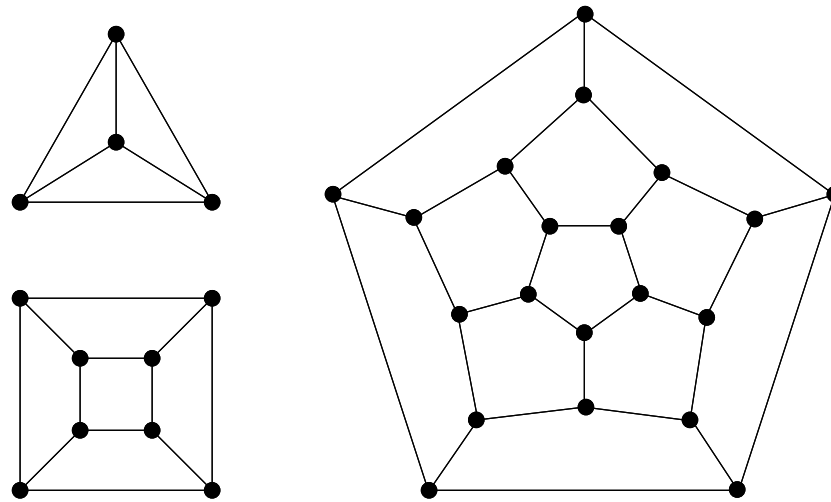


The protocol complexes of the standard 1-round wait-free layered immediate snapshot protocols for 2 and 3 processes.

Topology of the layered immediate snapshot protocol complexes II

Definition.

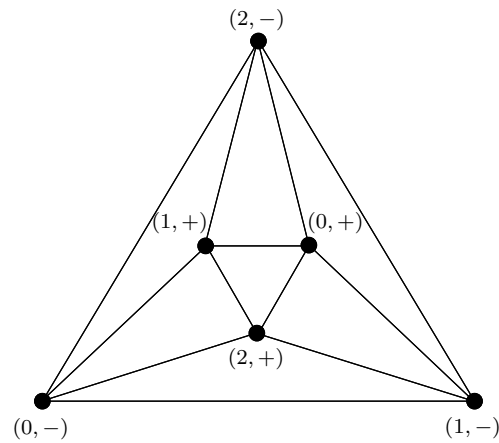
Schlegel diagrams



Schlegel diagrams of tetrahedron, cube and dodecahedron.

Iterating Schlegel diagrams

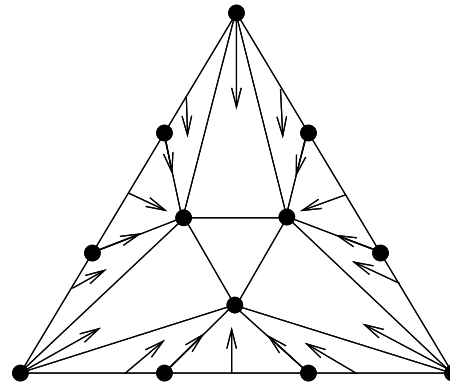
The standard chromatic subdivision can be obtained by iterating Schlegel diagrams of cross-polytopes.



Schlegel diagrams of cross-polytopes for $d = 1, 2$.

Explicit equivariant collapsing sequence

The top dimensional simplices of $\text{Ch } \Delta^n$ are indexed by ordered tuples of disjoint sets $(B_1, \dots, B_t)$ such that $B_1 \cup \dots \cup B_t = [n]$. The lower dimensional simplices of $\text{Ch } \Delta^n$ are indexed by pairs of tuples of non-empty sets $((B_1, \dots, B_t)(C_1, \dots, C_t))$, such that B_i 's are disjoint, and $C_i \subseteq B_i$ for all i .



2. Immediate snapshot protocol complexes

Topology of the immediate snapshot protocol complexes I

Definition.

A **witness structure** is a sequence of pairs of finite subsets of $\mathbb{Z}_+$, denoted $\sigma = ((W_0, G_0), \dots, (W_t, G_t))$, with $t \geq 0$, satisfying the following conditions:

- (P1) $W_i, G_i \subseteq W_0$, for all $i = 1, \dots, t$;
- (P2) $G_i \cap G_j = \emptyset$, for all $0 \leq i < j \leq t$;
- (P3) $G_i \cap W_j = \emptyset$, for all $0 \leq i \leq j \leq t$;
- (P4) the subsets $W_1, \dots, W_t$ are all nonempty.

- the set $W_0 \cup G_0$ is called the **support** of σ and is denoted by $\text{supp } \sigma$;
- the **ghost set** of σ is the set $G(\sigma) := G_0 \cup \dots \cup G_t$;
- the **active set** of σ is the complement of the ghost set

$$A(\sigma) := \text{supp } (\sigma) \setminus G(\sigma) = W_0 \setminus (G_1 \cup \dots \cup G_t);$$

- the **dimension** of σ is

$$\dim \sigma := |A(\sigma)| - 1 = |W_0| - |G_1| - \dots - |G_t| - 1.$$

Topology of the immediate snapshot protocol complexes II

Definition.

Given a function $\bar{r} : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+ \cup \{\perp\}$, we consider the set

$$\text{supp } \bar{r} := \{i \in \mathbb{Z}_+ \mid \bar{r}(i) \neq \perp\}.$$

This set is called the **support set** of $\bar{r}$.

A **round counter** is a function $\bar{r} : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+ \cup \{\perp\}$ with a finite support set.

Definition.

Assume $\bar{r}$ is a round counter. We define a simplicial complex $P(\bar{r})$ as follows:

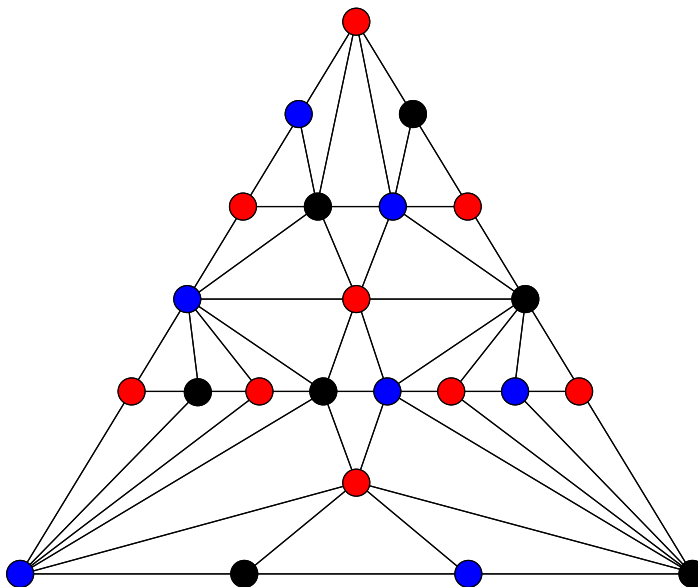
- the simplices of $P(\bar{r})$ are indexed by witness structures σ satisfying the following properties:
 - (1) $\text{supp } \sigma = \text{supp } \bar{r}$;
 - (2) for all $p \in A(\sigma)$, we have $|M(p, \sigma)| = r(p) + 1$;
 - (3) for all $p \in G(\sigma)$, we have $|M(p, \sigma)| \leq r(p) + 1$.
- the dimension of the simplex indexed by σ is $\dim \sigma$, its vertices are $\Gamma_{A(\sigma) \setminus \{a\}}(\sigma)$, where a ranges through the set $A(\sigma)$.

The complex $P(\bar{r})$ is called the **immediate snapshot complex** associated to the round counter $\bar{r}$.

Topology of the immediate snapshot protocol complexes III

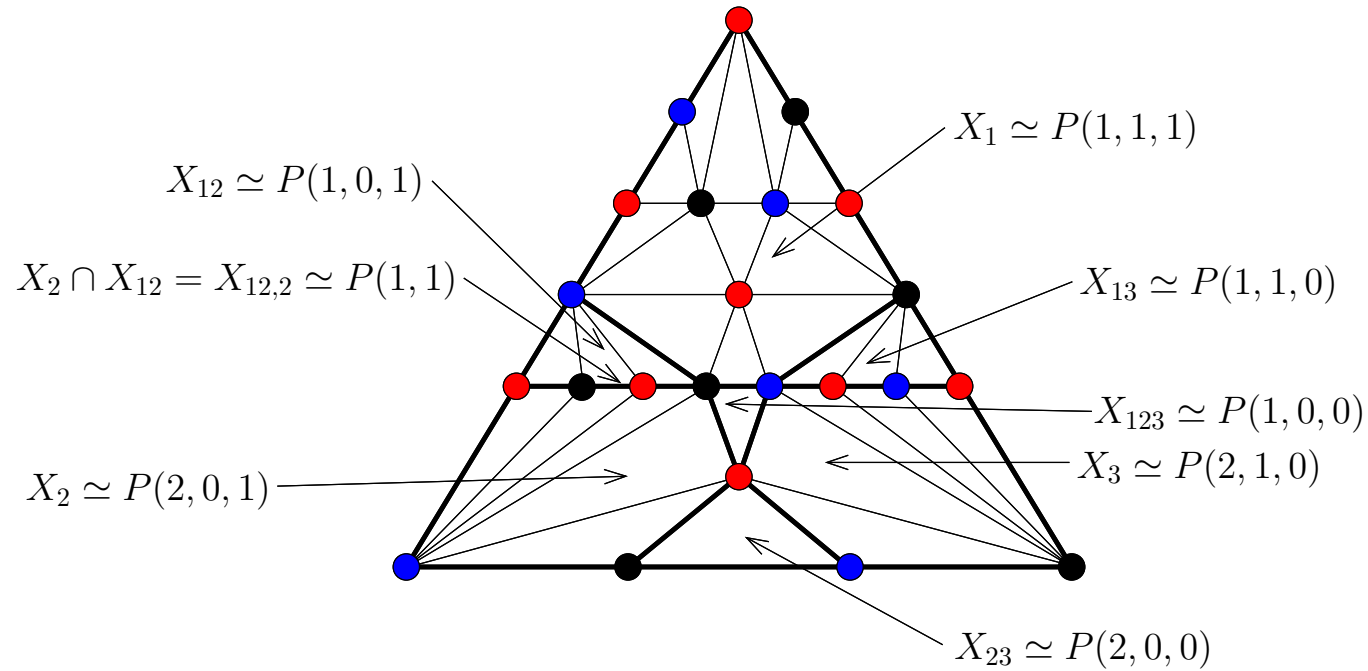
Theorem.

Protocol complexes for wait-free IS protocols for $n + 1$ processes for an arbitrary round counter are simplicially homeomorphic to an n -simplex.



The protocol complex of a wait-free IS protocol for 3 processes.

Topology of the immediate snapshot protocol complexes IV



The standard stratification of the protocol complex of a wait-free IS protocol for 3 processes.

3. Snapshot protocol complexes

Topology of the view complex I

Definition.

An n -**view** is a $2 \times t$ -matrix $\begin{pmatrix} V_1 & \cdots & V_{t-1} & [n] \\ I_1 & \cdots & I_{t-1} & I_t \end{pmatrix}$, where $t \geq 1$, and the entries are subsets of $[n]$ satisfying the following properties:

- $\emptyset \neq V_1 \subset \cdots \subset V_{t-1} \neq [n]$,
- the sets $I_1, \dots, I_t$ are disjoint,
- $\emptyset \neq I_k \subseteq V_k$, for $k = 1, \dots, t-1$.

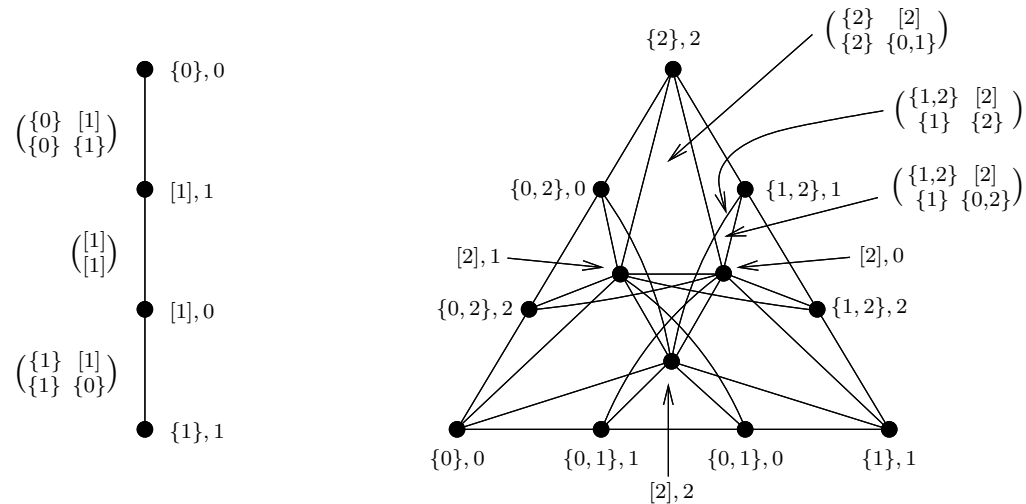
The abstract simplicial complex View^n is defined as follows:

- the set of vertices $V(\text{View}^n) := \{(V, x) \mid x \in V \subseteq [n]\}$;
- a subset $S \subseteq V(\text{View}^n)$ forms a simplex if and only if there exists an n -view W , such that for each $(V, x) \in S$ there exists $1 \leq k \leq t$, such that $V = V_k$ and $x \in I_k$.

Topology of the view complex II

Theorem.

Protocol complexes for 1-round wait-free read-write protocols of $n + 1$ processes are equivariantly collapsible to the iterated standard chromatic subdivision of an n -simplex.



The view complexes for 2 and 3 processes.

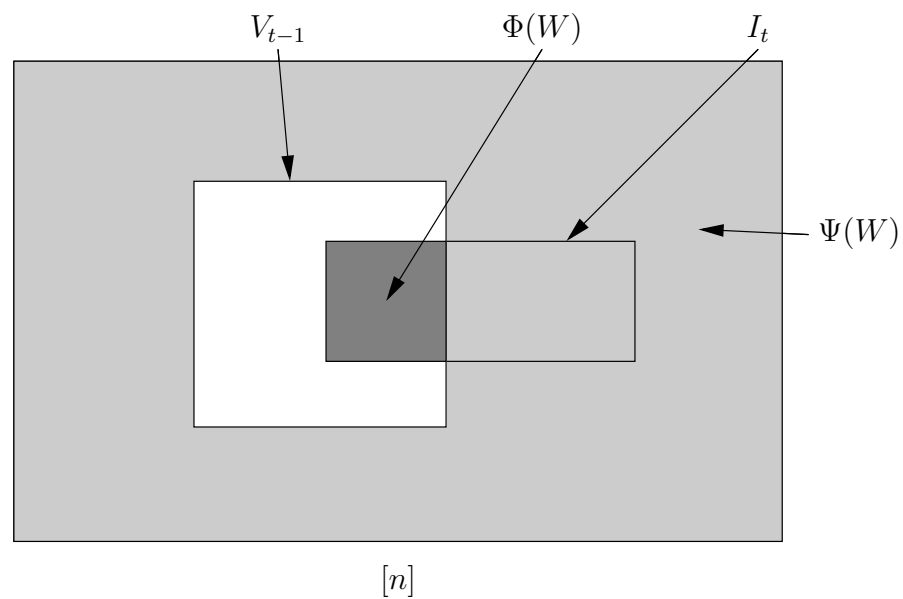
Topology of the view complex III

For an arbitrary n -view

$$W = \begin{pmatrix} V_1 & \cdots & V_{t-1} & [n] \\ I_1 & \cdots & I_{t-1} & I_t \end{pmatrix}$$

we set

$$\Phi(W) := \begin{pmatrix} V_1 & \cdots & V_{t-1} & [n] \\ I_1 & \cdots & I_{t-1} & I_t \cap V_{t-1} \end{pmatrix}, \quad \Psi(W) := \begin{pmatrix} V_1 & \cdots & V_{t-1} & [n] \\ I_1 & \cdots & I_{t-1} & I_t \cup ([n] \setminus V_{t-1}) \end{pmatrix}.$$



SUMMARY

- Simplicial methods are useful in theoretical distributed computing.
- Recent work provides rigorous foundation for this interdisciplinary area.
- Protocol complexes are configuration spaces encoding all possible executions of protocols.
- Combinatorial topology of protocol complexes is sophisticated, contains answers to distributed computing questions.

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M. Herlihy, D.N. Kozlov, S. Rajsbaum, *Distributed Computing through Combinatorial Topology*, 1st Edition, Morgan Kaufmann / Elsevier, 2014, 336 pp.

D.N. Kozlov, *Chromatic subdivision of a simplicial complex*, Homology, Homotopy and Applications **14** (2012), no. 2, 197–209.

D.N. Kozlov, *Topology of the view complex*, preprint 10 pages.
arXiv:1311.7283 [cs.DC]

D.N. Kozlov, *Topology of the immediate snapshot complexes*, preprint 24 pages.
arXiv:1404.5813 [cs.DC]

D.N. Kozlov, *Witness structures and immediate snapshot complexes*, 26 pages.
arXiv:1404.4250 [cs.DC]

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Lecture III:

Mathematics of the Weak Symmetry Breaking task

Plan of Lecture III:

1. Solvability of Weak Symmetry Breaking
2. Abstract simplex paths and their deformations
3. Fast protocols

1. Solvability of Weak Symmetry Breaking

Rank-Symmetric Protocols for Weak Symmetry Breaking

Weak Symmetry Breaking is a standard task specified by the following:

- The processes have no input values.
- Their output values are 0 and 1.
- The task is solvable if there exists an IS wait-free protocol such that in every execution in which all processes participate, not all processes decide on the same value.

In addition, this protocol is required to be *rank-symmetric*.

Definition.

Let $I_1, I_2 \subseteq \{0, \dots, n\}$, such that $|I_1| = |I_2|$, and let $r : I_1 \rightarrow I_2$ be the order-preserving bijection. Assume E_1 is an execution of the protocol, in which the set of participating processes is I_1 , and E_2 is an execution of the protocol, in which the set of participating processes is I_2 . Then, the protocol is called **rank-symmetric** if for every $i \in I_1$, the process with ID i must decide on the same value as the process with ID $r(i)$. The protocol will certainly be rank-symmetric, if each process only compares its ID to the ID's of the other participating processes, and makes the final decision based on the relative rank of its ID.

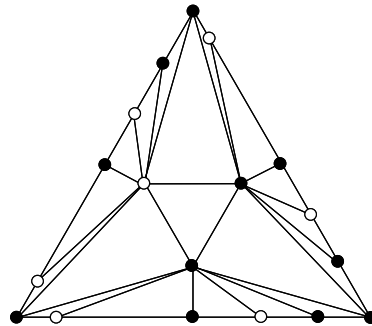
Mathematical Formulation of Solvability of WSB I

Definition.

A subdivision $\text{Div}(\Delta^n)$ of the standard n -simplex Δ^n is called **hereditary** if for all $I, J \subseteq [n]$ such that $|I| = |J|$, the linear isomorphism $\varphi_{I,J}$ is also an isomorphism of the subdivision restrictions $\text{Div}(\Delta^I)$ and $\text{Div}(\Delta^J)$.

A labeling $\lambda : V(\text{Div}(\Delta^n)) \rightarrow L$, where L is an arbitrary set of labels, is called **compliant** if for all $I, J \subseteq [n]$ such that $|I| = |J|$, and all vertices $v \in V(\text{Div}(\Delta^I))$, we have

$$\lambda(\varphi_{I,J}(v)) = \lambda(v).$$



An example of a hereditary subdivision with a compliant binary labeling.

Mathematical Formulation of Solvability of WSB II

Definition.

Given a subdivision $\text{Div}(\Delta^n)$, the labeling $\chi : V(\text{Div}(\Delta^n)) \rightarrow [n]$ is called a **coloring** if

- (1) any two vertices of $\text{Div}(\Delta^n)$, which are connected by an edge, receive different colors;
- (2) for any $v \in V(\text{Div}(\Delta^n))$, we have $\chi(v) \in V(\text{supp}(v))$.

A subdivision which allows a coloring is called **chromatic**.

Theorem.

Weak Symmetry Breaking for $n + 1$ processes is solvable in the immediate snapshot wait-free model if and only if there exists a hereditary chromatic subdivision $\text{Div}(\Delta^n)$ of Δ^n and a binary compliant labeling $b : V(\text{Div}(\Delta^n)) \rightarrow \{0, 1\}$, such that for every n -simplex $\sigma \in \text{Div}(\Delta^n)$ the restriction map $b : V(\sigma) \rightarrow \{0, 1\}$ is surjective.

Mathematical Formulation of Solvability of WSB III

Definition.

Assume $\text{Div}(\Delta^n)$ is a hereditary chromatic subdivision of the standard n -simplex and $b : V(\text{Div}(\Delta^n)) \rightarrow \{0, 1\}$ is a binary labeling. A simplex path $\Sigma = (\sigma_1, \dots, \sigma_l)$ is said to be in **standard form** if l is even, σ_1 and σ_l are 0-monochromatic, and $\sigma_2, \dots, \sigma_{l-1}$ are not monochromatic.

Theorem

Assume we are given a hereditary chromatic subdivision $\text{Div}(\Delta^n)$, a binary labeling $b : V(\text{Div}(\Delta^n)) \rightarrow \{0, 1\}$, and an n -dimensional geometric simplex path Σ in standard form. Then there exists a chromatic subdivision $\mathcal{S}(\text{Div}(\Delta^n))$ of the geometric simplicial complex $\text{Div}(\Delta^n)$, such that

- (1) only the interior of Σ is subdivided;
- (2) it is possible to extend the binary labeling b to $\mathcal{S}(\text{Div}(\Delta^n))$ in such a way that $\mathcal{S}(\Sigma)$ has no monochromatic n -simplices.

Index Lemma I

Let M be an orientable simplicial manifold with boundary ∂M . Let c be a coloring of the vertices of M . We say that a simplex is *properly colored* if all colors of its vertices are different. We want to *count simplices by orientation*, to which end we set

$$C(\sigma, c) := \begin{cases} 1, & \text{if } \sigma \text{ is properly colored and its color orientation is same as induced by} \\ & \text{the manifold orientation;} \\ -1, & \text{if } \sigma \text{ is properly colored and its color orientation is the opposite of} \\ & \text{the one induced by the manifold orientation;} \\ 0, & \text{if } \sigma \text{ is not properly colored.} \end{cases}$$

Definition.

The **content of** c in M is the number of simplices properly colored by c , counted by orientation:

$$C(M, c) := \sum_{\sigma \in M, \dim \sigma = n} C(\sigma, c).$$

Index Lemma II

Definition.

For each $i \in [n]$, the ***i*th index** of M is the number of $(n - 1)$ -simplices of ∂M , properly colored with values from $\text{Face}_i(\Delta^n)$, counted by orientation:

$$I_i(M, c) := \sum_{\tau \in \partial M, \dim \tau = n-1} C(\tau, c).$$

Index Lemma.

If M is an oriented pseudomanifold colored by c , then for each $i \in [n]$ we have the identity

$$C(M, c) = (-1)^i I_i(M, c).$$

The case when N is a prime power

If WSB is solvable, then there exists a compliant binary coloring, hence there is an $(n + 1)$ -coloring of $\text{Ch}^N \Delta^n$ with no monochromatic simplices.

On the other hand, we have the following fact.

Proposition.

There exist integers $k_0, k_1, \dots, k_{n-1}$, such that the number of monochromatic simplices in $\text{Ch}^N \Delta^n$, counted by orientation, is

$$(-1)^n + \sum_{q=0}^{n-1} k_q \binom{n+1}{q+1}.$$

The contradiction now follows from the fact that the binomial coefficients $\binom{n+1}{1}, \dots, \binom{n+1}{n}$ are all divisible with a prime p if $n + 1$ is a power of p .

The case when N is not a prime power

Castaneda - Rajsbaum Theorem.

WSB is solvable when N is not a prime power.

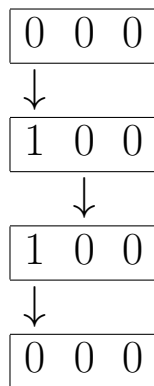
The idea is to start with a coloring with 0 content. This means that the number of monochromatic n -simplices is zero, counted with orientation. Then we link simplices of opposite orientation by simplex paths and use a quite sophisticated *path subdivision algorithm* to eliminate the monochromatic pair.

2. Abstract simplex paths and their deformations

Abstract Simplex paths

Definition.

Let k and n be positive integers. An **abstract simplex path** P is a triple (I, C, V) , where I is an $[n]$ -tuple of boolean values, C is a $(k - 1)$ -tuple of elements of $[n]$, and V is a $(k - 1)$ -tuple of boolean values. We request that the $(k - 1)$ -tuple C satisfies an additional “no back-flip” condition: for all $i = 1, \dots, k - 2$, we have $C_i \neq C_{i+1}$.



Cube loops

Definition.

An $[n]$ -**cube loop** is a directed cycle in $\text{Cube}_{[n]}$, which contains the origin, and does not have any self-intersections.

To describe an $[n]$ -cube loop one can start at the origin and then list the directions of the edges as we trace the loop. This procedure will yield a t -tuple $(q_1, \dots, q_t)$, with $q_i \in [n]$, for all $1 \leq i \leq t$.

Conversely, given such a t -tuple $Q = (q_1, \dots, q_t)$, it describes an $[n]$ -cube loop if and only if the following conditions are satisfied:

- (1) *the cycle condition*: for every $l \in [n]$, the number $\sharp(l, Q)$ is even;
- (2) *no self-intersections*: for all $1 \leq i < j \leq t$, such that $(i, j) \neq (1, t)$, there exists $l \in [n]$, such that the number $\sharp(l, Q[i, j])$ is odd.

Deformations of abstract simplex paths:

Vertex Expansion

Assume $P = (I, C, V)$ is an abstract simplex path of length $k \geq 3$ and dimension n . The input data for the vertex expansion consists of:

- a number m , such that $2 \leq m \leq k - 1$, called position of the expansion;
- an $[n]$ -tuple $D = (d_0, \dots, d_n)$ of boolean values;
- an $[n]$ -cube loop $(q_1, \dots, q_t)$, such that $q_1 = C_{m-1}$ and $q_t = C_m$.

We set $(c_0, \dots, c_n) := R^m(P)$.

Definition.

The **vertex expansion** of a path $P = (I, C, V)$ with respect to the input data (m, D, Q) is a new path $\tilde{P}$ of dimension n defined as follows:

- $I(\tilde{P}) = I$;
- $C(\tilde{P})$ is obtained from C by inserting the tuple $(q_2, \dots, q_{t-1})$ between C_{m-1} and C_m ;
- $V(\tilde{P})$ is obtained from V by replacing V_{m-1} with the tuple $w_1, \dots, w_{t-1}$, where for all $i = 1, \dots, t - 1$:

$$w_i = \begin{cases} c_{q_i}, & \text{if } \#(q_i, (q_1, \dots, q_i)) \text{ is even,} \\ d_{q_i}, & \text{if } \#(q_i, (q_1, \dots, q_i)) \text{ is odd.} \end{cases}$$

Cube p -paths

Definition.

Assume $p \in [n]$. An $[n]$ -**cube p -path** is a directed path I in the graph $\text{Cube}_{[n]}$, such that

- I starts at the origin, and ends in the vertex $(0, \dots, 0, 1, 0, \dots, 0)$, where 1 is in position p ;
- I does not have any self-intersections;
- I contains exactly one edge parallel to the p th axis.

In line with the situation of the $[n]$ -cube loops, the $[n]$ -cube p -path is given by a tuple $(q_1, \dots, q_t)$, where $q_1, \dots, q_t \in [n] \setminus \{p\}$, together with a number $1 \leq s \leq t-1$, subject to the following conditions:

- (1) *the path condition*: for every $l \in [n] \setminus \{p\}$, the number $\sharp(l, Q)$ is even;
- (2) *no self-intersections*: for all $1 \leq i < j \leq s$, there exists $l \in [n] \setminus \{p\}$, such that the number $\sharp(l, Q[i, j])$ is odd; the same is true for all $s+1 \leq i < j \leq t$.

Deformations of abstract simplex paths:

Edge Expansion

The input data for the edge expansion consists of:

- a number m , such that $2 \leq m \leq k - 2$;
- an $([n] \setminus \{C_m\})$ -tuple $D = (d_0, \dots, \widehat{d}_{C_m}, \dots, d_n)$ of boolean values;
- an $[n]$ -cube C_m -path given by $Q = (q_1, \dots, q_t)$, such that $q_1 = C_{m-1}$ and $q_t = C_{m+1}$, together with a number $1 \leq s \leq t - 1$.

Definition.

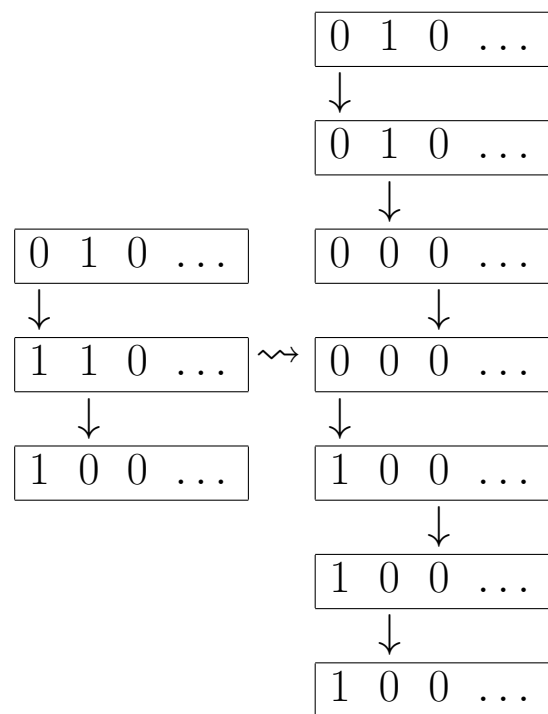
The **edge expansion** of a path P with respect to the input data (m, D, Q, s) is a new path $\tilde{P}$ of dimension n defined as follows:

- $I(\tilde{P}) = I$;
- $C(\tilde{P})$ is obtained from C by inserting the tuple $(q_2, \dots, q_s)$ between C_{m-1} and C_m , and then inserting the tuple $(q_{s+1}, \dots, q_{t-1})$ between C_m and C_{m+1} ;
- $V(\tilde{P})$ is obtained from V by replacing V_{m-1} with the tuple $w_1, \dots, w_s$, and then inserting the tuple $w_{s+1}, \dots, w_{t-1}$ between V_m and V_{m+1} , where for all $i = 1, \dots, t - 1$:

$$w_i = \begin{cases} c_{q_i}, & \text{if } \#(q_i, (q_1, \dots, q_i)) \text{ is even,} \\ d_{q_i}, & \text{if } \#(q_i, (q_1, \dots, q_i)) \text{ is odd.} \end{cases}$$

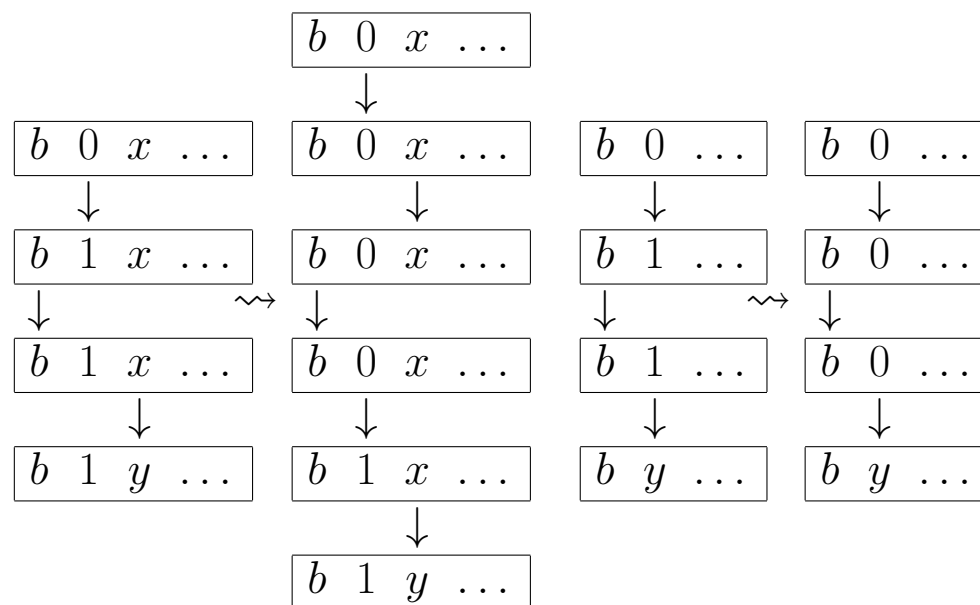
Deformations of abstract simplex paths:

The summit move



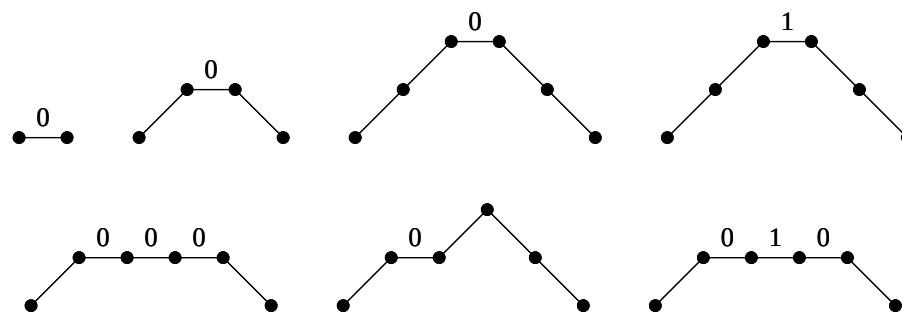
Deformations of abstract simplex paths:

The plateau move



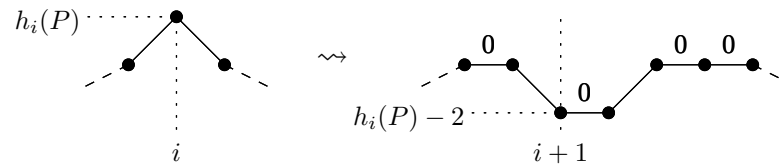
The generic and the special plateau moves.

Height graphs



Examples of the height graphs.

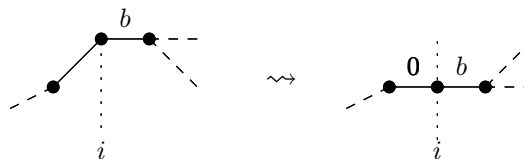
The height graph changes



The height graph change in the summit move.



The height graph change in a generic plateau move.



The height graph change in a special plateau move.

Exhaustive expansions

Let P be an abstract simplex path, and let $\tilde{P}$ be either a vertex expansion $P(m, D, Q)$ or an edge expansion $P(m, D, Q, s)$. We describe how to associate a $[1] \times [n]$ -array A to that pair $(P, \tilde{P})$. To start with, in any case we set $A[0, -] := R^m(P)$. Proceeding to the second row: for a vertex expansion $P(m, D, Q)$ we set $A[1, -] := D$, while for an edge expansion $P(m, D, Q, s)$ we set $A[1, i] := D_i$, for all $i \in [n] \setminus \{C_m\}$, and $A[1, C_m] = V_m$.

Since the vertices of $\text{Cube}_{[n]}$ are indexed by all functions $\alpha : [n] \rightarrow [1]$, the $[1] \times [n]$ -array of boolean values A can be used to associate boolean $[n]$ -tuples to vertices of $\text{Cube}_{[n]}$. Specifically, to a vertex $\alpha : [n] \rightarrow [1]$ of $\text{Cube}_{[n]}$ we associate the $[n]$ -tuple $A(\alpha) := (A[\alpha(0), 0], A[\alpha(1), 1], \dots, A[\alpha(n), n])$. Let $M(A)$ denote the set of all $\alpha : [n] \rightarrow [1]$, such that $A(\alpha)$ is monochromatic.

Definition.

Let the paths P and $\tilde{P}$ be as above. The expansion $\tilde{P}$ is called **exhaustive** if, there exists a partial matching on the vertices in $M(A)$, such that, for each unmatched $\alpha \in M(A)$, there exists i , such that for all $j \in [n]$, the numbers $\alpha(j)$ and $\sharp(j, (q_1, \dots, q_i))$ have the same parity. In the case when $\tilde{P}$ is an edge expansion, we require in addition that if $\alpha(C_m) = 0$, then $1 \leq j \leq s$, and if $\alpha(C_m) = 1$, then $s + 1 \leq j \leq t$.

The subdivision algorithm

Use the summit and plateau moves to reduce the height function until $\max_i h_i(P) = 1$.

Deal with this much simplified path directly.

3. Fast protocols

SUMMARY

- WSB leads to the rich theory of subdivisions of the simplicial paths where much structure remains mysterious.
- The explicit protocol solving WSB (when it is solvable) remains outside of reach.
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